

Course Title: Machine Learning and Data Visualization

Course Code	404								
Course Title	Machine Learning and Data Visualization								
Credits	4								
Course Category	Major Course								
Level of Course	300-399 (Higher Level)								
Teaching per Week	2 Hours Theory + 4 Hours Practical work								
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)								
Review / Revision	-								
Implementation Year:	A.Y. 2025-2026								
Purpose of Course	This course aims to provide in-depth knowledge of advanced supervised and unsupervised machine learning algorithms. Students will gain hands-on experience in classification, clustering, and model optimization techniques using Python. The course emphasizes practical applications, real-world datasets, and interpretable model building. By the end, learners will be capable of executing and deploying end-to-end machine learning solutions.								
Course Objective	Ob-1: To provide a comprehensive understanding of advanced classification algorithms such as Decision Trees, Random Forest, KNN, and Naïve Bayes. Ob-2: To introduce students to unsupervised learning techniques like K-Means, Hierarchical Clustering, and DBSCAN, along with their practical applications. Ob-3: To develop competency in model optimization through cross-validation methods, hyperparameter tuning, and regularization techniques. Ob-4: To explain the importance of feature selection and guide students in using effective feature selection methods in machine learning models. Ob-5: To enable students to carry out end-to-end machine learning projects using real-world datasets and deploy their models using tools such as Flask and Streamlit.								
Pre-requisite	A basic understanding of supervised learning algorithms, data preprocessing, and evaluation metrics is essential. Prior knowledge of Python programming and libraries such as Pandas, NumPy, and Scikit-learn is recommended.								
Course Outcomes	CO1: Understand and explain the working principles of advanced supervised and unsupervised machine learning algorithms. CO2: Apply appropriate classification or clustering techniques to solve real-world problems using structured datasets. CO3: Analyze model performance using cross-validation, tuning, and evaluation metrics to improve prediction accuracy. CO4: Evaluate feature importance and apply regularization and feature selection techniques to optimize models. CO5: Design, implement, and deploy complete machine learning solutions using real-world case studies and Python tools.								
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8
	CO1								
	CO2								
	CO3								
	CO4								
	CO5								

Course Content	Unit-1: Advanced Supervised Learning – Classification Algorithms 1.1 Decision Trees 1.1.1 Concept and Working of Decision Trees 1.1.2 Gini Index, Entropy, and Information Gain 1.1.3 Overfitting and Pruning in Decision Trees 1.2 Random Forest 1.2.1 Introduction to Ensemble Learning 1.2.2 Bagging Concept 1.2.3 Random Forest Algorithm 1.2.4 Feature Importance in Random Forest 1.2.5 Use-case: Predicting Customer Churn using Python 1.3 K-Nearest Neighbors (KNN) 1.3.1 Distance Metrics (Euclidean, Manhattan) 1.3.2 Choosing the Value of K 1.3.3 Strengths and Weaknesses of KNN 1.4 Model Optimization 1.4.1 Hyperparameter Tuning 1.4.2 Cross-validation in Practice 1.4.3 Understanding Overfitting and underfitting 1.5 Support Vector Machines (SVM) 1.5.1 Concept of Hyperplanes and Margins 1.5.2 Linear vs Non-Linear Classification 1.5.3 Kernel Trick and Common Kernels (RBF, Polynomial) 1.5.4 Pros and Cons of SVM Unit-2: Unsupervised Learning – Clustering Techniques 2.1 Introduction to Unsupervised Learning 2.1.1 Use Cases and Applications 2.1.2 Challenges of Unsupervised Learning 2.2 K-Means Clustering 2.2.1 Algorithm Steps 2.2.2 Choosing the Number of Clusters (Elbow Method) 2.2.3 Limitations of K-Means 2.2 DBSCAN (Density-Based Clustering) 2.3.1 Concepts of Core, Border, and Noise Points 2.3.2 Parameters: eps and min_samples 2.3.3 Comparison with K-Means 2.5 Model Optimization and Tuning 2.5.1 Cross-Validation Techniques with Hyperparameter Search 2.5.2 K-Fold Cross-Validation 2.5.3 Stratified K-Fold 2.5.4 Leave-One-Out Cross-Validation UNIT 3: Real-World Applications and Deployment 3.1 Case-Based Learning 3.1.1 Predictive Analytics in Healthcare 3.1.2 Credit Risk Analysis in Banking 3.1.3 Movie Recommendation Engine (Collaborative Filtering) 3.2 Model Deployment Basics 3.2.1 Saving and Loading Models (Pickle and Joblib) 3.2.2 Using Flask for Creating Python APIs 3.2.3 Simple Model Deployment on Localhost 3.3 Introduction to MLOps 3.3.1 Basics of MLOps and its Importance 3.3.1.1 Understanding MLOps(Machine Learning Operations) 3.3.1.2 Importance in real-world ML project lifecycle 3.3.1.3 Comparison of traditional ML vs MLOps lifecycle 3.3.1.4 Key benefits: automation, reproducibility, monitoring
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	3.4 Case Discussion: End-to-End ML Pipeline 3.4.1 Case Study: A spam detection model for an email system 3.4.2 Data ingestion (daily incoming emails) 3.5 Applied Machine Learning and Case Studies 3.5.1 End-to-End ML Project Workflow 3.5.2 Problem Definition to Deployment 3.5.3 Data Splitting, Preprocessing, Modeling, Evaluation 3.6 Working with Real-World Datasets 3.6.1 UCI and Kaggle Datasets Use Cases 3.6.2 Basics of Text and Image Classification 3.7 Model Evaluation and Interpretation 3.7.1 Confusion Matrix 3.7.2 ROC Curve and AUC 3.7.3 Precision-Recall Trade-off UNIT 4: Data Visualization and Storytelling 4.1 Principles of Effective Data Visualization 4.1.1 What Makes a Visualization Effective 4.1.2 Understanding the audience 4.1.2.1 Clarity, accuracy, and simplicity 4.1.3 Storytelling with visuals 4.1.3 Types of Charts and When to Use Them 4.1.3.1 Bar chart, Line graph, Histogram 4.1.3.2 Pie chart, Box plot, Heatmap 4.1.3.3 Tree map, Scatter plot, Bubble chart 4.2 Power BI and Tableau for Business Intelligence 4.2.1 Overview and comparison of Tableau vs Power BI 4.2.2 Installing and setting up Power BI Desktop and Tableau Public 4.3 Connecting to datasets using PowerBI: Excel, CSV, Web, SQL Server 4.3.1 Building Visual Reports 4.3.2 Designing dashboards with multiple visuals 4.3.3 Creating calculated fields and DAX (Power BI) 4.3.4 Filters, slicers, and drill-through capabilities 4.4 Creating interactive dashboards with user input using PowerBI 4.4.1 Publishing and Sharing Reports using PowerBI 4.4.2 Sharing links and embedding dashboards [Suggested Practical work : Students will create a mini-project on following during Practical hours: 1) Sales Dashboard: Analyze sales performance of a company 2) Weather Data Visualization: Plot temperature trends across cities 3) AQI Dashboard: Visualize Air Quality across metro cities 4) Education Insights: Analyze pass rates, dropout trends, etc. Final Expected Deliverables are: i) Cleaned dataset ii) Python-based EDA report iii) BI tool dashboard iv) Short presentation explaining the story behind the data]
Reference Books	1) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Aurélien Géron, ISBN: 978-1098125974 2) Pattern Recognition and Machine Learning, Springer, Christopher M. Bishop, ISBN: 978-0387310732 3) Python Machine Learning, Packt Publishing, Sebastian Raschka, Vahid Mirjalili, ISBN: 978-1801819312 4) Data Science for Business, O'Reilly Media, Foster Provost, Tom Fawcett, ISBN: 978-1449361327 5) Storytelling with Data: A Data Visualization Guide for Business Professionals, Wiley, Cole Nussbaumer Knaflitz, ISBN: 978-1119002253

	6) Machine Learning, McGraw Hill Education India, Tom M. Mitchell (Adapted Edition), ISBN: 978-9339222724 7) Data Visualization and Dashboarding, BPB Publications, Ajoy Kumar Ray, ISBN: 978-9391392971 8) Artificial Intelligence and Machine Learning, PHI Learning, Pratap K. Dixit, ISBN: 978-9391818518 9) Data Science and Machine Learning using Python, BPB Publications, Srinivas Reddy, ISBN: 978-9389898752 10) Business Intelligence and Analytics using Power BI, Wiley India, G. Nagarjuna, Ramesh Babu, ISBN: 978-8126578914
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.